WASP-43b

R-0428

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_250108` · created 2026-07-19T10:03:35+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460683.8513 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.057
Transit depth (Rp/Rs)^2	3.84 +/- 2.22 [%]
Orbital Inclination (inc)	86.79 +/- 2.96
Airmass coefficient 1 (a1)	1.6 +/- 0.27
Airmass coefficient 2 (a2)	-0.25 +/- 0.24
Scatter in the residuals of the lightcurve fit is	18.85 %
Best Comparison Star	#3 - [168, 66]
Optimal Aperture	4.43
Optimal Annulus	11.95
Transit Duration (day)	0.0608 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.057 vs 0.1594 (deviation 0.64σ)
Duration fit vs published	0.0608 d vs 0.0512 d (deviation 1.74σ)
Mid-time O–C	-3.7 min
Depth SNR	3.4
Residual scatter	18.85 %

Light curve

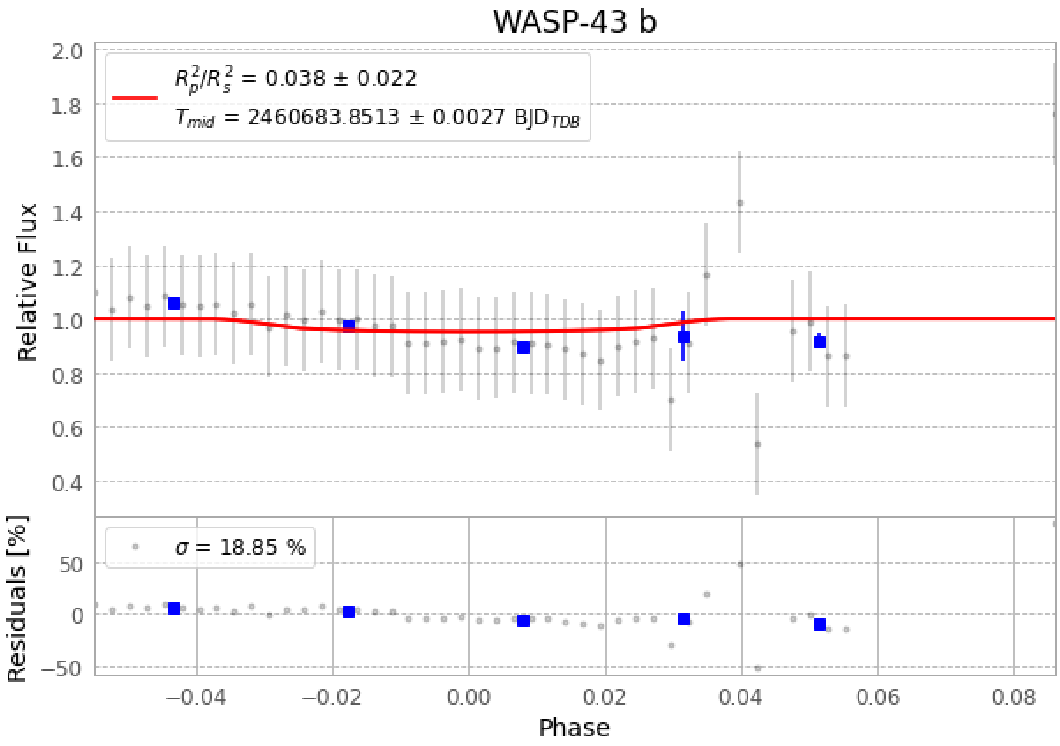

Figure 1 — FinalLightCurve_WASP-43 b_2025-01-08.png (SHA-256 14b24aa2fa06b1a4…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [463, 192], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 db20421de2b36bee…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

55 FITS frames (36 MB), `WASP-43250108071515.FITS` → `WASP-43250108100016.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250108.log (ce60fad445a5…), FinalLightCurve_WASP-43 b_2025-01-08.png (14b24aa2fa06…), FinalLightCurve_WASP-43 b_2025-01-08.csv (0fd25c2ac2b0…)

Compute resources

Wall time 255.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)